Sean Zhao

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Education

University of Toronto Toronto, ON

Bachelor of Applied Science in Computer Engineering *Expected Apr. 2029*

- GPA of 3.97/4.00, Dean's Honour List, and the ECE Top Student Award, given to the top 3 students of the year
- Relevant coursework in C programming, C++ data structures and algorithms, and digital systems (last two in progress)

Experience

Data Security Developer Intern May 2026 – Aug. 2026

1Password *Toronto, ON*

- Migrated the Rust cryptographic layer that protects user data in the 1Password app to a new set of algorithms, and was responsible for choosing the algorithms, designing the migration, and writing the implementation
- Benchmarked the candidate algorithms on Android, iOS, and ARM devices and confirmed that the new implementation stayed within 3% of the old one's performance
- Moved all 21 legacy call sites to the new algorithms behind feature flags, with the app able to read data written under either algorithm, so existing records stayed readable and no user had to reset or re-encrypt anything
- Wrote automated tests covering backward compatibility and rollback for the migration
- **Received the Intern Trailblazer Award** for innovation and adopting new technologies

Projects

Pilot | *TypeScript, JavaScript, Playwright, MCP* Sept. 2026

- **Won The Token Company's LLM Price-Saving Challenge at HackMIT 2026** with a tool that lets apps and coding agents pull data from websites that have no official API, using simple shared commands such as jobs.search
- Built a Playwright agent that writes a JavaScript driver for each website and tests it before saving it, so later requests run the saved driver without calling an AI model again and Pilot can repair the driver when the site changes
- Connected the drivers to Codex and Claude Code through Model Context Protocol (MCP), which cut token use by 50% in an app-building test compared with running the same task without Pilot

GameNet | *Rust, Tokio, QUIC, TLS 1.3, ML-KEM* Jan. – Sept. 2026

- Built a Rust tunnel that lets friends join a self-hosted game server with their normal game clients by connecting to a public relay, which forwards each player to the host over its own QUIC stream
- Secured the connection between the host and the relay with TLS 1.3, certificate and hostname checks, and a required X25519MLKEM768 key exchange that combines classical and post-quantum cryptography
- Added port claims that expire, limits on connections, and host tokens that can be rotated or recovered if stolen
- Measured a median reply time of 44–47 ms through GameNet compared with 213–215 ms through Pinggy, a similar TCP tunnel, in a synthetic latency test

Alias | *Next.js, FastAPI, Ghidra, LLM4Decompile, Gemini, Docker, Modal* Jan. 2026

- **Won Best Cybersecurity Hack (1Password track) at UofTHacks 13** with a tool that explains what an executable does and flags security risks before anyone runs it
- Built the analysis pipeline, which decompiles the binary with Ghidra, rewrites the output as readable C with LLM4Decompile, and has Gemini write a plain-English summary, with the models running in containers on Modal

Awards

CyberTitan Nationals (2025): 1st place overall

CyberPatriot National Finals (2025): Cisco 2nd place and 6th overall

Canadian Computing Competition: Junior perfect score (2023) and Senior Distinction (2024, 2025)

Technical Skills

Languages: Rust, C, C++, Python, TypeScript, JavaScript, Java

Systems & Backend: Linux, Docker, Git, Tokio, QUIC, TLS 1.3, Node.js, Express.js, FastAPI

Web & Data: React, Next.js, PostgreSQL, MongoDB, OpenCV, MediaPipe

AI & Security: MCP, Playwright, Modal, Ghidra, applied cryptography